

	SESSION 1 (Morning) 25th September 2025 (09:00- 12:00) PHYSICAL ATTENDANCE OF LOCAL PARTICIPANTS IS MANDATORY @ The Open University of Sri Lanka, Colombo, Sri Lanka				
	08.00 – 9.00 REGISTRATION				
	09:00– 09:30 Inauguration - Auditorium – Faculty of Health Sciences				
	PARALLEL SESSION 1	PARALLEL SESSION 2	PARALLEL SESSION 3	PARALLEL SESSION 4	PARALLEL SESSION 5
	Theme: Plastic Pollution and Health Impacts: Microplastics in Terrestrial & Marine Ecosystems	Theme: Technological Innovations in Sustainable Environment	Theme: Climate Action, Computational Modelling and Environmental Sustainability	Theme: AI and Smart Technologies for Monitoring and Restoring Ecosystem Health	Theme: Technological Innovations in Sustainable Environment & Climate Action, Computational Modelling and Environmental Sustainability
Hall	A – Seminar Room – Faculty of Natural Science	B – Research Lab - Department of Electrical & Computer Engineering – Faculty of Engineering Technology	C – Faculty Board Room - Faculty of Engineering Technology	D - Faculty Board Room - Faculty of Natural Science	E – Auditorium – Faculty of Health Sciences
	Chair: Prof. Hasintha Wijesekara/ Prof. Uthpala Jayawardana	Chair: Prof. T.M. Pallewatta/ Prof. D.A.R. Dolage	Chair: Prof. Kumari Fernando/ Prof. Thrikawala	Chair: Prof. Ayantha Gomez/ Dr. Menaka Ranasingha	Chair: Prof. Shiromi Karunaratne/Prof. Buddhika Aruggoda
	Co-Chair/Zoom Co-Host: Dr. Lakshini Fernando	Co-Chair/Zoom Co-Host: Dr. Ruminda Wimalasiri	Co-Chair/Zoom Co-Host: Dr. Nadeera Gunathilaka	Co-Chair/Zoom Co-Host: Dr. Maneesha Caldera	Co-Chair/Zoom Co-Host: Dr. Nadeeka Miguntanna
Zoom Link	https://learn.zoom.us/j/92406207870?pwd=IAZ4FnJFgyWkawQpWPGnl6RZNRcily.1	https://learn.zoom.us/j/92657944426?pwd=8sD3ELy5VdBJjJ5K1b4UJ6q7j5YVAr.1	https://learn.zoom.us/j/94452358059?pwd=fGddHcQDBUBxncrVb0L6xscz9Ettwi.1	https://learn.zoom.us/j/95940534439?pwd=Hi9DVxk97g3b95dpigTpPKK5ie7arv.1	https://learn.zoom.us/j/99717032502?pwd=0bwgdU0FUjljdSFSyeZZOwmwLIQxK.1

	PARALLEL SESSION 1	PARALLEL SESSION 2	PARALLEL SESSION 3	PARALLEL SESSION 4	PARALLEL SESSION 5
09.30 – 09.45	ID 8 Investigation of microplastics in commercially available salt in Sri Lanka	ID 11 Design and fabrication of sustainable flat washers using textile fabric waste	ID 27 Restoring Sri Lanka’s Central Highland Cloud Forests in Bopaththalawa: A Nature-Based Fiscal Strategy for Climate Adaptation and Water Security	ID 37 Detecting Underwater Macroplastic Pollution: A Review on AI-Based Computer Vision Methods and Their Relevance to Sri Lanka’s Marine Ecosystem	ID 29 Unrefined montmorillonite nanoclay-cinnamon wood biochar composite for simultaneous removal of hardness and fluoride in groundwater
09:45 – 10:00	ID 9 Effect of regional water on microplastic release from tea bags in Sri Lanka	ID 33 Closed-Loop Copper Recovery: Integrating Electrolytic with Circular Economy Practices	ID 52 Analysis of Cadmium and Chromium Temporal Variations in Muthurajawela Wetland: A Heavy Metal Contamination Assessment from Kerawalapitiya Waste Park	ID 38 AI and IoT-Driven Framework for Monitoring and Restoring of Mangrove Ecosystem Health in Coastal Sri Lanka	ID 28 Feasibility Study on Forecasting Water Level Fluctuation of the Senanayaka Samudra Reservoir, Sri Lanka
10:00 – 10:15	ID 34 ALDFG and Single-Use Plastics Based Marine Pollution Prevention in Sri Lanka: A Case Study from Rockland Corporate Responsibility	ID 36 The Mechanical Properties of PET Fiber Reinforced Concrete	ID 56 Designing a Wildlife Crossing Structure to Mitigate the Impacts of the Central Expresway Stage 4 - Kurunegala to Dambulla	ID 39 Early Detection and Diagnosis of Coconut Leaf Diseases in Sri Lanka’s Coconut Triangle: A Systematic Review	ID 40 Study on Impact of Railway Bridge Piers on Flood Inundation at Adjacent Riverbanks of Kelani River, Sri Lanka
10:15 – 10:30	ID 53 Microplastic Emission Dynamics in an Urban River under Rainfall Events	ID 44 Development of a Sustainable and Skin-Friendly Marigold Flower-Inspired Resort Wear Collection Using Natural Dyeing Techniques	ID 57 Contribution of Mobile Sources Emissions to Ambient Air Quality: Case Study on the Baseline Road, Greater Colombo, Sri Lanka	ID 42 Plastic Waste Leakage in Urban River Basins of Sri Lanka: A Geospatial and Machine Learning Perspective	ID 31 Solidified Water Treatment Sludge as Potential Supplementary Cementitious Material in Pervious Concrete Pavers for Mitigating Heat Island Effect
10:30 – 10:45	ID 60 Microfiber Removal in Wastewater Treatment: A Stage-wise Analysis from Sri Lanka	ID 48 Microgrid for Low Voltage Distribution Systems and a Model for Optimum Power Sharing	ID 41 Extending the European Union Deforestation-free Regulation to Sri Lanka’s Land-Based Shrimp Aquaculture: Environmental, Regulatory and Trade Justifications	ID 59 A Systematic Review of Clinical Strategies for Bloodless Surgery: A Machine Learning Perspective	ID 47 Green Synthesized Sustainable Filter Material for Enhanced Wastewater Treatment
10.45 – 11.00	ID 77 Microfiber Shedding in Domestic Laundering Using Washing Machines: What Does Multivariate Analysis Reveal?	ID 49 A Comprehensive analysis of grid power quality issues in the proposed electrification of Panadura - Veyangoda	ID 51 Temporal Variations in Leachate Contamination Risk Assessment at Kerawalapitiya Waste Park within Muthurajawela Wetland, Sri Lanka: An LPI-Based Investigation	ID 61 Towards a Socio-Techno-Ecological Framework (RESCUE) for Smart Waste Management for Urban Areas in Sri Lanka: A Conceptual Model	ID 62 An investigation of the influence of microplastics on the thermal conductivity of seawater
11:00 – 11:15	ID 93 Occurrence of microplastics in selected liquid oral medications and intravenous infusions in Sri Lanka	ID 64 Upcycled Sustainable Evening Wear for Sri Lankan Women: Inspired by the Elegance of the Turkey Tail Mushroom	ID 65 Revising and Optimizing Iso-Yield Curves for the Diverse Agro-Climatic Zones of Sri Lanka	ID 71 UTAUT-Based Predictive Analysis of Youth Adoption of Gamified Environmental Sustainability Apps	ID 106 Review of the Heavy Metal Filtration Efficiency of Mangrove Root Systems
11:15 – 11:30	ID 95 Cytogenotoxic effects of polyethylene terephthalate (PET) microplastics on Allium cepa root tip cells: implications for abnormal cell proliferation	ID 104 Advancing Sustainability through Material Innovation and Circular Designs in Plastics	ID 45 Integrating Biodiversity Assessments as a Complementary Framework for EUDR Implementation in High-Risk Commodity Landscapes	ID 74 IoT-Enabled Real-Time Water Quality Monitoring for Shrimp Farming: Trends, Challenges, and Future Directions in Sri Lanka	ID 96 Conversion of Softlogic Furniture Factory to a Net-Zero Energy Facility
11.30 – 11.45		ID 75 Convertible Ladies’ Casual Wear Collection (S/S) for the U.S. Inspired by the Pearl Oyster	ID 30 Design a Retro-Reflective eco-friendly clay roof tile to reduce the Urban Heat Island (UHI) effect	ID 84 Comprehensive Study of Trends in Autonomous Vehicle	ID 87 Design and Implementation of a Low-Power Edge-WSN Architecture for Mangrove Health Monitoring in Restoration Sites
11.45 – 12.45	Lunch				

	SESSION 2 (Evening) 25th September 2025 (13:00- 16:00) PHYSICAL ATTENDANCE OF LOCAL PARTICIPANTS IS MANDATORY @ The Open University of Sri Lanka, Colombo, Sri Lanka			
	PARALLEL SESSION 1	PARALLEL SESSION 2	PARALLEL SESSION 3	PARALLEL SESSION 4
	Theme: Policy and Regulatory Frameworks for Waste Management	Theme: Technological Innovations in Sustainable Environment	Theme: Climate Action, Computational Modelling and Environmental Sustainability	Theme: Public Awareness and Educational Initiatives
H all	A – Seminar Room – Faculty of Natural Science	B – Research Lab - Department of Electrical & Computer Engineering – Faculty of Engineering Technology	C – Faculty Board Room - Faculty of Engineering Technology	D - Faculty Board Room - Faculty of Natural Science
	Chair: Prof. Sanath Wijeyasingha/ Prof. Aruna Shantha	Chair: Prof. Chathura Herath/ Prof. Uditha Rathnayaka	Chair: Dr. Ibrahim Mohamed/ Dr. Gehan Anthonys	Chair: Prof. Thiwankee Wickramasinghe/Prof. S. Pathmanesan
	Co-Chair/Zoom Co-Host: Mr. W.D. Manoj	Co-Chair/Zoom Co-Host: Dr. Nandika Miguntanna	Co-Chair/Zoom Co-Host: Dr. Chamila Dias	Co-Chair/Zoom Co-Host: Dr. Athula Samarakoon
Zoom Link	https://learn.zoom.us/j/92406207870?pwd=IAZ4FnJFgyWkawQpWPGnl6RZNRClcy.1	https://learn.zoom.us/j/92657944426?pwd=8sD3ELy5VdBJjJ5K1b4UJ6q7j5YVAr.1	https://learn.zoom.us/j/94452358059?pwd=fGddHcQDBUBxnCrVb0L6xscz9Ettwi.1	https://learn.zoom.us/j/95940534439?pwd=Hi9DVxk97g3b95dpigTpPKK5ie7arv.1

	PARALLEL SESSION 1	PARALLEL SESSION 2	PARALLEL SESSION 3	PARALLEL SESSION 4
13:00 – 13:15	ID 21 Plastic Waste Policies and Informal Waste Pickers: A Policy Gap Analysis in Sri Lanka	ID 68 Sustainable Braided Beach Resort Wear for Sri Lanka: Natural Fabrics and Eco-Friendly Dyeing Innovations	ID 78 Spatial and Temporal Patterns of Riverine Plastic Waste Inputs to the Ocean in Sri Lanka: Integrating Waste Management Archetypes and Hydrological Drivers	ID 4 The Missing Link in Sustainability: Cultivating Eco-Love as a Tool for Behaviour Change in Plastic Use
13:15 – 13:30	ID 22 Onsite Construction Waste Generation in Sri Lanka: Patterns, Causes, and Policy Implications	ID 69 A Low-Cost and Power-Efficient Customizable IoT Platform that Facilitates Large-Area Outdoor Remote Sensing via Packet Relaying	ID 79 Modeling Microplastic Impacts on Soil Quality in Sri Lanka Using Data Analytics and Graph Data Science	ID 5 A theory synthesis and systematic literature review on cognitive biases in climate shaped plastic behaviors
13:30 – 13:45	ID 24 Assessment of Municipal Solid Waste in Kaduwela, Sri Lanka: Composition Analysis and Policy Implications for Waste Management	ID 70 A Simplified Vehicle Speed Estimation Algorithm on FPGA	ID 80 Microplastics and Water Reserves in Sri Lanka: A Spatial Hotspot and Risk Assessment Framework	ID 6 Promoting Pro-Environmental Behavior in Sri Lanka: A Literature-Based Conceptual Framework Integrating Psychology and Cultural Values
13:45 – 14:00	ID 66 Bridging the Legal Gaps in Plastic Waste Governance in Sri Lanka’s Regulatory Framework	ID 76 Soil respiration response to green-synthesized metallic nanoparticles assessed via CO ₂ trapping with NaOH	ID 81 Using Graph Data Science to Assess the Impact of Plastic Pollution on Air Quality in Sri Lanka	ID 10 Understanding the Continued Use of Plastic Bags among the Adults over 30 years old in Sri Lanka: Patterns, Impacts, and Alternatives
14:00 – 14:15	ID 35 Repurposing Plastic Waste into Circular Production: A Model for Sustainable Textile Manufacturing	ID 98 Enhancing the autonomous emergency landing zone detection system for drones in fail safe-mode when in signal lost situation.	ID 58 Multi-Temporal GIS-Based Analysis of Surface Water Susceptibility to Pollution in the Kala Oya Basin, Sri Lanka	ID 13 Relationship between Environmental, Social and corporate performance: Sri Lankan context from 2017 to 2021
14:15 – 14:30	ID 12 Alternatives for plastics -Challenges and Opportunities	ID 101 Performance Evaluation of Glass Fiber Reinforced Pervious Concrete for Urban Infrastructure	ID 85 GIS Hotspot Risk Scoring Model for Plastic Leakage in Urban Lakes (Sri Lanka)	ID 67 Eco-driven community education for sustainable plastic use: Colombo case study
14:30 – 14:45		ID 102 Towards Sustainable Laboratory Practices: Eco-Friendly Treatment of Decentralized Wastewater	ID 90 Analyzing Shoreline Data to Assess the Erosion at Bolifushi through Drone Imagery: Towards Sustainable Coastal Planning	ID 88 From Waste to Wardrobe: Awareness Occasion Wear Collection Crafted from Recycled Plastics, Fabric Scraps, and Leftover Materials inspired by Goddess Gala
14:45 – 15:00		ID 103 Analysis of Chemical Properties of Washed and Sieved Offshore Sand	ID 94 Bias Correction of GCM Simulated Rainfall Data for Sri Lanka: Using Statistical Methods Across Different Climate Zones and Seasonal Regimes	ID 89 Influence of Rewards and Gamified Experience in Waste Management Among School Students in Jaffna College Sri Lanka
15:00 – 15:15		ID 7 Comparative Evaluation of Bioplastic Films from Invasive Water Hyacinth and Banana Peel Waste: A Sustainable Alternative to Plastics for Circular Economy Integration	ID 99 Groundwater Modelling to Investigate the Oil Spill Contamination of the Chunnakam Aquifer in Jaffna Peninsula	ID 91 Technical Sustainability as a Crucial Determinant of Long-Term Sustainability in Community-Based Water Supply Schemes in Sri Lanka
15.15-15.30			ID 105 Clay biochar composites in treating laboratory wastewater	ID 46 Plastic Pollution and Protecting the Right to a Clean and Healthy Environment in Sri Lanka
15.30 – 16.30	Awarding & Closing Ceremony - Auditorium – Faculty of Health Sciences			
	Refreshment			